HACKHARVARD - FINAL BUILD PLAN

ReefTrace

Pollution source intelligence for visible plastic: moving objects use trajectory-conditioned inference, stationary objects use source-to-deposition simulations, and both can return Unknown.

MOVING + STATIONARY	3 SOURCE ZONES	TRANSPORT MODEL + UNKNOWN

Scientific boundary: ReefTrace demonstrates dual-mode attribution among known tank source zones. Real geographic attribution requires a separate site model and field validation.

Version 2.0 | Source-first final 36-hour scope



Contents

The document is organized around the physical build, the measurement and inference stack, the hardware-software boundary, and the evidence needed for a credible live demonstration.

1. Executive summary 3

2. Final scope and claim boundaries 4

3. What judges will see 5

4. Physical system and hardware design 6

5. Software architecture 9

6. End-to-end data flow 12

7. Calibration and software-hardware interfaces 13

8. Math, inference, and scientific limitations 15

9. Data model and database schema 18

10. Dashboard design 21

11. Sponsor integrations with claim discipline 23

12. Thirty-six-hour build plan 24

13. Team roles 26

14. Test and validation plan 27

15. Live demo script 29

16. Fallback scope ladder 31

17. Risks and mitigations 31

18. Presentation assets 32

19. Roadmap after the hackathon 33

20. Final definition of done 33

1. Executive summary

ReefTrace is a controlled proof of concept for **plastic-pollution source intelligence**. A fixed overhead camera watches a shallow tank containing tap water and artificial coral. A circulation pump creates a repeatable surface current. Three labeled inlets represent possible pollution-entry zones. Computer vision detects each visible plastic object, converts image coordinates to physical tank coordinates, and classifies the observation as **moving**, **stationary/deposited**, or **uncertain**.

Moving plastic and stationary plastic require different evidence. If the object is moving, ReefTrace conditions an ensemble of backward transport simulations on the observed positions and velocity vectors. If the object is already stuck on the artificial coral, ReefTrace cannot reverse a nonexistent trajectory; instead, it runs or queries forward simulations from each candidate inlet over a range of possible release times and compares how often simulated particles arrive in the observed deposition region. Both pathways produce ranked candidate sources, an evidence explanation, an uncertainty measure, and **Unknown / insufficient evidence** when the result is not defensible.

The tank's measured surface-current vector field drives the transport model. Repeated physical releases are used as held-out validation of the model, not as a requirement for manually teaching every possible path. The outside product would replace the tank field with a site-specific digital twin built from mapped boundaries, candidate source zones, time-varying current data, and wind, tide, or wave inputs where relevant. Deployment still requires appropriate local data and validation, but not a new library of manual plastic releases for every detected object.

An above-water XY rover and suction nozzle provide a secondary response layer. After a human approves collection, the rover removes the detected plastic and the camera verifies the result. Photon can optionally carry an approval request through a supported messaging channel, Butterbase can store observations and outcomes, and Chalk can serve derived context features. None of those services calculates the scientific source score or bypasses local safety controls.

The scientifically honest claim is narrow and useful:

> *In a calibrated tabletop tank with three known candidate inlets, ReefTrace can use measured water movement to rank likely entry points for both moving and stationary visible plastic, abstain when evidence is weak, and optionally trigger human-approved collection.*

The prototype **does not** establish the geographic origin of real marine debris. Real-world attribution would require geographically referenced field observations, time-varying current and wind data, a model appropriate to the material, uncertainty analysis, and field validation.

One-sentence pitch

> *ReefTrace is a pollution-forensics system: it combines camera observations with a site transport model to estimate where moving or stationary plastic most likely entered, then uses robotic collection as an optional response.*

Thirty-second pitch

Ocean cleanup is valuable, but cleanup alone is reactive and already well explored. ReefTrace focuses on the harder question: **where is the plastic entering the environment?** In a controlled reef tank, an overhead camera observes visible plastic and a measured current field powers a local transport model. Moving objects are analyzed with trajectory-conditioned backward simulations. Stationary objects are analyzed with forward source-to-deposition simulations. ReefTrace ranks three known inlets, explains the evidence, and returns Unknown when it cannot distinguish them. A human-approved rover can then collect the object, but collection demonstrates the response loop rather than defining the product. The prototype validates tank-scale inlet attribution; field-scale geographic attribution remains a separate deployment and validation problem.

The complete loop

Observe -> Classify motion state -> Model transport -> Rank source zones -> Explain uncertainty -> Optionally collect -> Verify -> Learn

2. Final scope and claim boundaries

In scope

- Tap water in a shallow, clear tank.
- Artificial coral below the surface as a visual habitat model.
- A fixed overhead camera with a full view of the tank.
- An above-water rover, preferably an XY gantry, that carries a surface suction nozzle.
- Visible floating plastic pieces large enough to detect and safely capture.
- One adjustable circulation pump operated at a fixed, calibrated setting during a run.
- Three fixed, known inlet locations.
- OpenCV detection, tracking, motion-state classification, tank-coordinate calibration, position vectors, velocity vectors, and measured surface-flow vectors.
- A site transport model that numerically moves virtual plastic through the measured tank current field.
- A moving-plastic pathway using observed trajectories and backward ensembles.
- A stationary-plastic pathway using forward simulations from candidate inlets to the observed deposition region.
- Source estimation among the three known inlets, with an evidence-based confidence or relative match score and an Unknown option.
- Physical releases used as validation trials, not as a required path-teaching procedure for every deployment.
- Human-approved physical collection and post-action visual verification.
- Python, FastAPI, Next.js, and Arduino as the core stack.
- Optional, adapter-based use of Chalk, Photon, and Butterbase.

Explicitly out of scope

- Charles River water, data, or claims.
- Real coral or any living coral organism.
- An underwater robot or submerged electronics.
- Sediment, dissolved pollutants, oil, fishing gear, submerged trash, or microplastics.
- Detection of plastic chemistry or polymer type.
- Open-ocean or real geographic source attribution.
- Attribution to a company, person, watershed, or polluter.
- Full computational fluid dynamics.
- A location-independent model that works without local boundaries, current information, candidate source zones, or validation.
- A claim that stationary debris has one uniquely recoverable origin when multiple sources can reach the same deposition area.
- Autonomous collection without a human approval and safety gate.

Language the team should use

Say: **"The model estimates which of three known tank inlets best explains the available evidence. Moving objects contribute trajectory evidence; stationary objects contribute endpoint and deposition evidence."**

Do not say: **"The AI identifies where ocean plastic came from."**

Say: **"The displayed number is a calibrated confidence or relative match score, the inference mode is visible, and the system can abstain."**

Do not say: **"A 78% score proves Inlet B caused the pollution."**

Say: **"This is a controlled demonstration of the sensing, inference, approval, collection, and verification loop."**

Say: **"The transport model is generated from local environmental inputs. Physical releases test the model; they are not the production source database."**

Do not say: **"The system never needs site-specific data or field validation."**

Product transfer beyond the tank

ReefTrace is designed around a reusable inference engine plus a site-specific environmental package. A new deployment does not begin by manually releasing plastic from every possible source. It begins by configuring:

- the modeled water boundary and coordinate system;
- candidate source zones such as inlets, outfalls, canals, or river mouths;
- time-indexed surface currents at suitable spatial resolution;
- wind, tide, wave, and diffusion terms when they materially affect floating plastic;
- mapped habitat, shoreline, and likely trapping regions;
- material and drift assumptions for the plastic category; and
- uncertainty, data-coverage, and release-time limits.

The system then precomputes or updates transport ensembles automatically. Live camera or sensor observations query that model through the moving or stationary branch. Independent observations, drifters, or documented events are still required to validate accuracy before the output is treated as geographic evidence. If the environmental data is missing or too coarse, the product must report that the site is unsupported rather than manufacture precision.

3. What judges will see

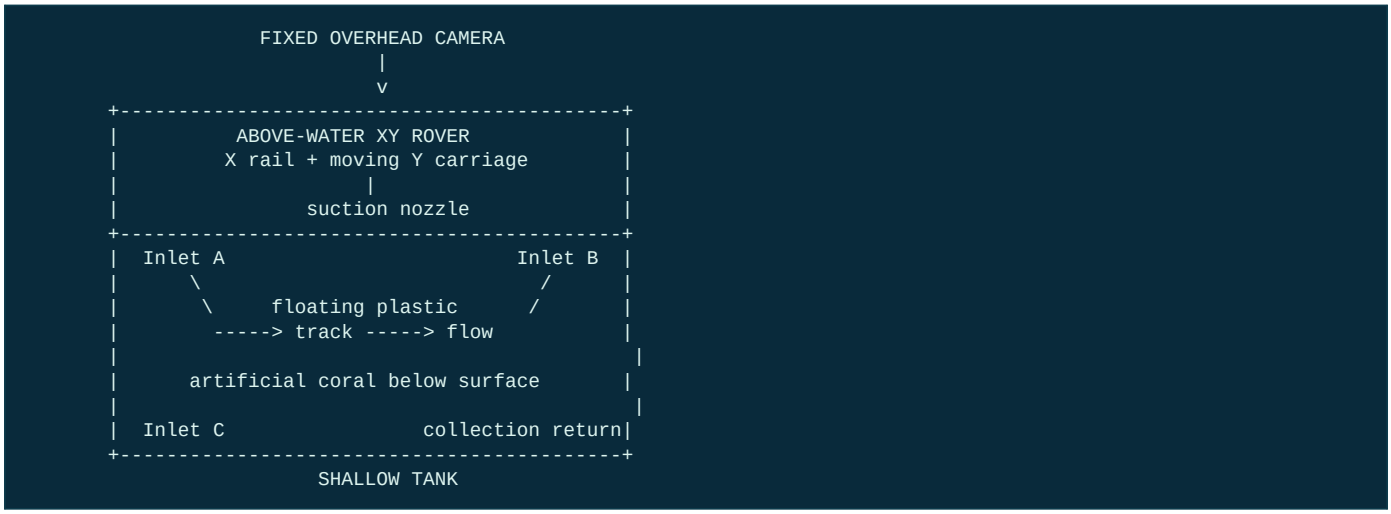
1. A clear tank contains tap water, submerged artificial coral, a gentle measured current, and three labeled candidate inlets.
2. The dashboard shows a transport map generated by numerically releasing virtual particles from the three inlets through the measured current field.
3. In the moving scenario, a judge secretly selects an inlet and releases one visible plastic piece.
4. The overhead camera detects the piece, classifies it as moving, and draws its observed trajectory in tank coordinates.
5. ReefTrace conditions backward simulations on the observed positions and velocities, then ranks the three inlets or displays Unknown.
6. In the stationary scenario, the camera detects a piece already stuck in a mapped artificial-coral deposition zone.
7. ReefTrace classifies it as stationary and compares forward source-to-deposition ensembles from all three inlets over a release-time window.
8. The dashboard displays the inference mode, ranked sources, evidence components, uncertainty, and the reason for any abstention.
9. The team reveals the known tank source for a validation run and shows whether the model was correct, wrong, or uncertain.
10. As a secondary finale, a teammate approves collection and the Arduino-controlled rover captures the object in a visible mesh chamber.
11. The camera verifies the result and the dashboard records the observation, model version, estimate, approval, and outcome.

4. Physical system and hardware design

4.1 Recommended layout

Use a clear tank approximately 40 x 30 cm with 8-12 cm of tap water. Place artificial coral on the bottom while preserving an open surface corridor for the plastic. Fix the camera to the frame above the tank; do not mount it on the moving carriage. A fixed camera makes the pixel-to-tank calibration stable. Mount the collection nozzle on an XY gantry above the water, with only the nozzle tip and tubing near the surface.

The tank is primarily an instrumented source-attribution testbed. The camera, stable geometry, known inlets, mapped coral obstacles, and measured current field matter more than a mechanically elaborate collector. If time forces a tradeoff, keep the transport model and both inference modes and simplify the collection mechanism to a fixed nozzle.



The circulation pump and collection pump have different jobs:

- **Circulation pump:** creates the repeatable tank current used in calibration and testing.
- **Collection pump:** creates suction at the nozzle. A mesh chamber upstream of this pump captures the plastic, and the water returns to the tank.

The circulation pump supports the scientific model. The collection pump supports the optional response demonstration.

4.2 Detailed bill of materials

Prices are planning ranges only, not current quotes. Use borrowed or lab parts wherever possible. Verify voltage, current, tubing diameter, and motor-driver compatibility before purchasing.

Subsystem	Item and target specification	Qty.	Planning range	Required?
Tank	Clear shallow tank, about 40 x 30 x 15 cm	1	$15-$30	Yes
Habitat	Artificial coral, weighted and fully submerged	2-3	$10-$25	Yes
Vision	1080p USB webcam, manual or lockable focus preferred	1	$20-$50	Yes
Lighting	Diffused LED bars or lamps, neutral white	2	$15-$30	Strongly recommended
Reflection control	Matte black backdrop and optional polarizing sheet/filter	1 set	$5-$25	Optional
Motion	Small XY gantry or repurposed plotter frame	1	$40-$100	Preferred
Motors	NEMA 17 or kit-compatible steppers	2	Often included	Yes for gantry

Subsystem	Item and target specification	Qty.	Planning range	Required?
Drivers	A4988, DRV8825, or kit-compatible stepper drivers with cooling	2	$8-$20	Yes for gantry
Controller	Arduino Uno, Nano, or compatible board	1	$12-$30	Yes
Limit sensing	Normally closed limit switches with brackets	3-4	$5-$15	Yes for gantry
Collection pump	12 V self-priming diaphragm or peristaltic pump, roughly 1-4 L/min	1	$18-$40	Yes
Pump switching	Logic-level MOSFET module with flyback protection	1	$5-$12	Yes
Capture	Transparent inline chamber with removable mesh basket	1	$8-$20	Yes
Plumbing	Food-safe silicone tubing, clamps, barbs, 10-15 mm intake nozzle	1 set	$10-$25	Yes
Flow	Adjustable low-voltage aquarium circulation pump	1	$12-$30	Yes
Flow markers	Reusable high-contrast floating discs or beads, retrieved after calibration	20-40	$5-$15	Yes
Inlets	Three funnels or tubes fixed at known surface coordinates	3	$5-$15	Yes
Test objects	Clean visible plastic fragments, about 5-15 mm, counted and recovered	10-20	$0-$5	Yes
Power	Listed 12 V DC supply sized for motors and pumps, plus 5 V buck if needed	1	$20-$40	Yes
Protection	Inline fuse holder/fuses, latching emergency stop, terminal blocks	1 set	$15-$35	Yes
Structure	Aluminum extrusion, plywood, acrylic, clamps, fasteners	1 set	$20-$60	Yes
Splash control	Clear splash shield and drip tray	1 set	$10-$25	Yes
Computer	Laptop with USB ports, Python, and Node.js	1	Existing	Yes

Expected cash total: about **$200-$450** if the team buys the tank, gantry, pumps, electronics, and frame. A borrowed plotter/gantry, webcam, and power supply can cut the total substantially.

4.3 Mechanical design notes

- Keep the camera fixed and place the gantry outside its critical line of sight when possible.
- Set the nozzle tip 2-5 mm above or barely at the water surface. Avoid pushing it deeply underwater.
- Use a wide intake that can accept the largest test piece without clogging.
- Place the removable mesh chamber **before** the collection pump.
- Return collected water gently at a location that does not create an uncalibrated jet in the observation region.
- Define a software no-go margin around tank walls, coral protrusions, inlet fixtures, and the return tube.
- Add physical end stops even if limit switches are present.
- Cover belts, leadscrews, and pinch points that judges may reach.
- Use a rigid frame. A camera or rail that shifts after calibration invalidates coordinate mapping.

4.4 Assembly sequence

Phase A - Tank and frame

1. Place the empty tank in a spill tray on a level table.
2. Install the rigid overhead frame and verify it cannot tip into the water.
3. Attach artificial coral to the tank bottom. Keep every decoration below the surface.
4. Mark three inlet locations on the rim and measure their coordinates from the tank origin.
5. Install the circulation pump and aim it along the long axis, not directly at a wall or coral branch.

Phase B - Camera and lighting

1. Mount the camera approximately normal to the water surface.
2. Frame the entire usable surface plus four calibration markers.
3. Lock focus and exposure if the camera supports it.
4. Mount diffused lights symmetrically. Add a dark surround or polarizer if reflections remain severe.
5. Tape or bolt the camera mount so it cannot move after calibration.

Phase C - Rover and collection loop

1. Assemble and square the XY gantry before positioning it over the tank.
2. Install X, Y, and optional redundant end-stop switches.
3. Attach the lightweight nozzle and secure the tubing so it cannot drag the carriage.
4. Connect nozzle -> mesh chamber -> collection pump -> gentle return line.
5. Manually test for leaks and collection with the gantry unpowered.
6. Mount the return line and verify that turning the collection pump on does not displace the target before capture.

Phase D - Electronics

1. Put the Arduino, motor drivers, MOSFET, fuses, connectors, and power distribution inside a raised splash-resistant enclosure.
2. Connect the Arduino to the laptop by USB.
3. Route motor and pump power from the fused 12 V supply through the emergency stop.
4. Connect the pump through the MOSFET driver, never directly to an Arduino pin.
5. Use the driver manufacturer's wiring and current-limit procedure for the selected stepper motors.
6. Use normally closed limit-switch wiring so a broken wire is detected as a fault.
7. Label every connector and provide strain relief and drip loops.

4.5 Electrical, water, and operating safety

This is a prototype around water. Safety outranks the demo.

- Power the table from a GFCI/RCD-protected outlet.
- Prefer listed low-voltage DC pumps. Keep mains-voltage equipment away from the tank.
- Put all exposed connectors and electronics above the maximum splash line and behind a clear shield.
- Add drip loops to every cable and tube approaching the tank.
- Fuse the DC supply near its output and size the fuse for the wiring and expected load.
- Use a latching emergency stop that removes motor and collection-pump power while leaving the laptop alive for diagnostics.
- Ground the frame and power supply according to their manufacturers' instructions.

- Never change wiring with power applied. Dry hands before touching controls.
- Inspect tubing, clamps, chamber seals, and the spill tray before each run.
- Keep towels and a nonconductive absorbent pad nearby; do not place a power strip on the floor.
- Stop immediately on leak, unexpected motor movement, jam, missing limit switch, overheating driver, or unstable frame.
- Add software travel limits, but do not rely on software as the only protection.
- Keep hands away from belts, leadscrews, and the rover during motion.
- Count and recover every calibration marker and plastic test piece. Do not pour plastic into a drain.
- Use clean tap water and disinfect or dry the system after the event. Do not use real environmental water.

4.6 Hardware acceptance gate

Do not connect autonomy until a teammate can safely perform all of the following from a manual test panel:

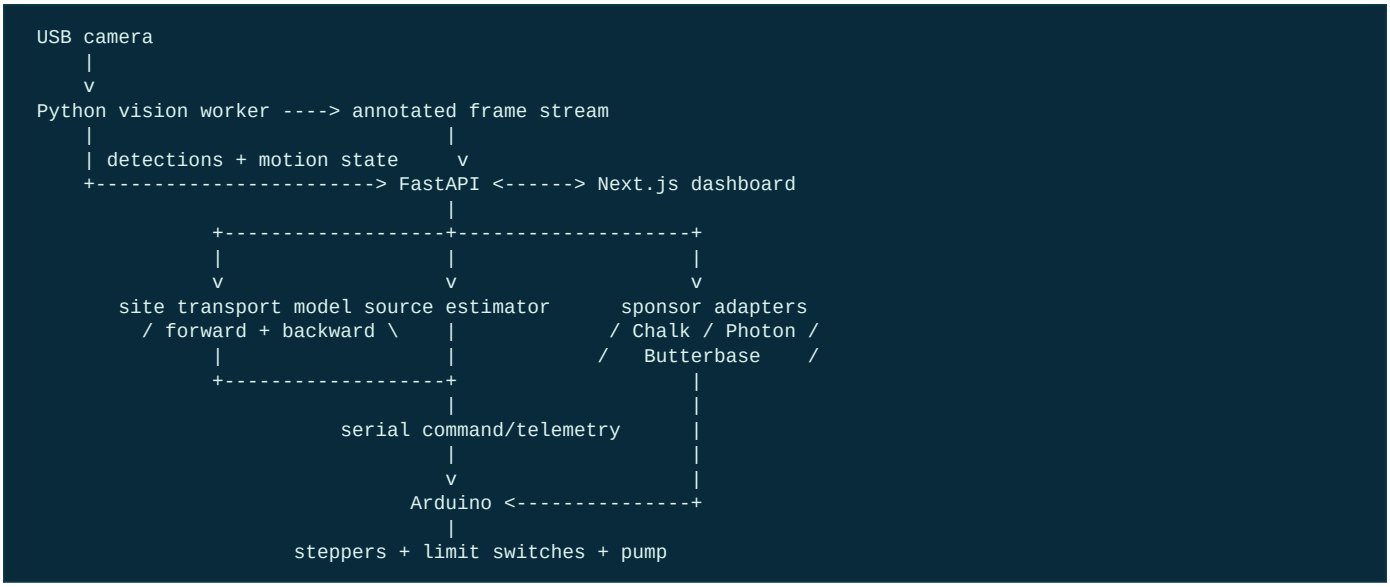
1. Trigger the emergency stop and confirm motor and pump power are removed.
2. Home both axes repeatably.
3. Move to all four safe corners without collision.
4. Click a point in the camera image and place the nozzle within 10 mm of that tank location.
5. Turn suction on for a bounded duration and capture a test piece.
6. Confirm the mesh chamber retains the object and the return flow stays controlled.
7. Rehome after a simulated serial disconnect or limit-switch fault.

5. Software architecture

5.1 Core stack

Layer	Technology	Responsibility
Vision and math	Python + OpenCV + NumPy/SciPy	Calibration, detection, motion classification, flow vectors, transport ensembles, source scoring
API and orchestration	FastAPI	State machine, APIs, WebSocket events, approvals, database adapters, hardware commands
Dashboard	Next.js + TypeScript	Live view, overlays, run controls, evidence, approval, history, safety state
Device control	Arduino firmware	Homing, bounded motion, limit switches, pump timing, fault reporting
Persistence	Butterbase adapter, with local SQLite/JSON fallback	Runs, tracks, estimates, approvals, actions, outcomes
Feature context	Optional Chalk adapter	Serve or log derived features used by the decision layer
Messaging	Optional Photon adapter	Deliver a human-readable approval request and receive a response where supported

5.2 Architecture diagram



The scientific critical path is local and reproducible:

Camera -> Python/OpenCV -> motion-state classifier -> site transport model -> source estimator -> evidence and uncertainty

The optional response path begins only after inference:

Source estimate -> human approval -> Arduino -> collection -> visual verification

Sponsor services are adapters around that loop, not single points of failure.

5.3 Runtime components

Vision worker

- Captures frames at a fixed resolution and timestamp.
- Undistorts the frame if camera calibration is available.
- Applies the tank homography.
- Masks the valid surface area and static obstacles.
- Detects high-contrast plastic using color/brightness thresholds first.
- Tracks objects using centroid association plus a Kalman filter or OpenCV tracker.
- Estimates track quality from visibility, size consistency, missed frames, and jitter.
- Classifies motion state as moving, stationary/deposited, or uncertain using filtered speed, displacement, and observation duration.
- Emits position, velocity, bounding box, motion state, and confidence metadata.

Use a simple, controlled detector before attempting a learned object detector. The test pieces can be intentionally bright and unlike the coral. A model is optional only if the threshold detector demonstrably fails.

Site transport-model worker

- Observes multiple reusable floating markers during calibration runs.
- Tracks their tank-coordinate velocities.
- Bins observations into a coarse spatial grid.
- Uses robust averages and interpolation to form the surface vector field.
- Stores the field with pump setting, water depth, camera calibration ID, obstacle map, and timestamp.

- Numerically releases ensembles of virtual particles from each candidate inlet.
- Supports backward integration conditioned on a moving observed track.
- Supports forward integration to mapped coral and deposition zones for stationary observations.
- Perturbs release time, local velocity, and diffusion parameters so the output is a distribution rather than one perfect line.

Source estimator

- Rejects tracks that are too short or low quality.
- Routes moving objects to trajectory-conditioned backward ensembles.
- Routes stationary objects to forward source-to-deposition ensembles.
- Compares simulated outcomes with the observed trajectory or endpoint.
- Produces candidate scores, evidence metrics, and quality flags.
- Returns one inlet only when fit and separation thresholds are met; otherwise returns Unknown.

Physical labeled releases are reserved for held-out validation of these simulated pathways. They are not the production inference database.

Action state machine

Recommended states:

```
BOOT -> SAFE_IDLE -> HOMING -> READY -> OBSERVING -> CLASSIFYING
     -> MOVING_INFERENCE or STATIONARY_INFERENCE -> ESTIMATING
     -> AWAITING_APPROVAL -> MOVING -> SUCTION -> VERIFYING
     -> COMPLETE -> READY

Any state -> FAULT -> SAFE_IDLE after explicit reset and rehome
```

The backend, not the UI or an external messaging service, owns the authoritative state. Every command includes a run ID, action ID, expiry time, and expected current state.

Arduino firmware

- Starts with outputs off.
- Requires a successful home before accepting absolute moves.
- Enforces hard travel bounds and maximum speed.
- Debounces limit switches.
- Limits pump-on duration.
- Reports position, switch state, pump state, and faults.
- Stops motion on malformed command, watchdog timeout, limit hit, or emergency stop.

5.4 Repository layout

```
reeftrace/
  apps/
    dashboard/          # Next.js UI
    api/                # FastAPI routes and orchestration
  services/
    vision/             # OpenCV capture, detection, tracking
    transport/          # current field, particle ensembles, deposition map
    estimation/         # motion routing, inlet scoring, uncertainty
    hardware/           # serial client and coordinate transforms
  firmware/
    reeftrace_arduino/  # homing, motion, switches, pump
  adapters/
    chalk/
    photon/
    butterbase/
  config/
    tank.yaml
    calibration.json
  tests/
    unit/
    integration/
    replay/
  data/
    sample_runs/        # short prerecorded demo/replay fixtures
```

Keep secrets in server-side environment variables and out of the browser, firmware, repository, screenshots, and prerecorded demos.

6. End-to-end data flow

1. **Load site model:** FastAPI creates `run_id`, loads the active calibration and transport-model versions, verifies the current-field profile, and checks camera and hardware health.
2. **Acquire frame:** The vision worker timestamps a frame using the host monotonic clock.
3. **Detect:** OpenCV segments the high-contrast plastic object inside the valid observation mask.
4. **Track:** The tracker associates the detection with an active `track_id` and filters jitter.
5. **Transform:** The homography converts pixel centroid `(u, v)` to tank coordinates `(x_cm, y_cm)`.
6. **Differentiate:** The estimator derives object velocity `(vx, vy)` from filtered positions.
7. **Classify motion:** After a minimum observation window, the system labels the object `moving`, `stationary`, or `uncertain`.
8. **Route the evidence:** Moving objects go to backward trajectory inference. Stationary objects go to forward source-to-deposition inference. Uncertain objects continue observing or return Unknown.
9. **Run ensembles:** The transport engine perturbs release time, velocity-field error, and diffusion, then simulates many plausible paths for every candidate inlet.
10. **Estimate:** It calculates candidate scores, evidence coverage, uncertainty, fit and separation checks, and Unknown status.
11. **Explain:** The dashboard shows inference mode, ranked sources, model assumptions, score type, and any abstention reason.
12. **Plan optional collection:** If the team elects to remove the object, the planner forecasts a moving interception point or uses the stationary endpoint, then checks reachability and obstacle margins.
13. **Request approval:** The dashboard displays the evidence and collection plan. Photon may mirror the request if the verified integration supports the chosen channel.
14. **Approve or deny:** The backend validates that the action has not expired and that the target and safety state remain valid.
15. **Move and collect:** FastAPI converts tank coordinates to gantry millimeters. Arduino executes a bounded move and suction pulse or reports a fault.

16. **Verify:** OpenCV records `captured`, `not_captured`, or `uncertain`.
17. **Persist:** The local store and Butterbase adapter record the observation, motion state, model version, ensemble summary, source estimate, action, approval, and outcome.
18. **Evaluate:** For labeled validation trials, the hidden source is revealed and compared with the estimate. Production observations do not require a known answer.

7. Calibration and software-hardware interfaces

7.1 Coordinate systems

Use explicit names and units:

- **Image frame:** pixels, origin at image top-left, axes `(u_px, v_px)`.
- **Tank frame:** centimeters, origin at the inside lower-left usable corner, axes `(x_cm, y_cm)`.
- **Gantry frame:** millimeters or motor steps from the homed mechanical origin, axes `(x_g, y_g)`.

Never send raw image pixels directly to the Arduino.

7.2 Camera-to-tank calibration

Place four or more fixed fiducial markers at known tank coordinates. Estimate a planar homography `H`:

```
s [x_cm, y_cm, 1]^T = H [u_px, v_px, 1]^T
```

After dividing by scale `s`, the software obtains physical tank coordinates. Validate the transform at at least five points not used to fit it. Record mean and maximum error. Recalibrate after any camera, tank, or frame movement.

Target: mean mapping error under 5 mm and maximum error under 10 mm across the safe collection area.

7.3 Tank-to-gantry calibration

Use at least three non-collinear targets visible to the camera. Move the nozzle to each target in manual mode and record gantry coordinates. Fit an affine transform:

```
[x_g, y_g, 1]^T = G [x_cm, y_cm, 1]^T
```

If the mechanism is square and aligned, `G` may be close to a scale plus offset. Still fit and validate it rather than assuming perfect alignment.

7.4 Surface-flow calibration

1. Fill to a marked water depth.
2. Set the circulation pump to the chosen fixed setting.
3. Wait for the flow to stabilize.
4. Release several reusable high-contrast floating markers from varied locations.
5. Track them for 30-60 seconds per trial.
6. Repeat at least five times while retrieving all markers after each trial.
7. Convert their positions and velocities to tank coordinates.
8. Aggregate observations into a coarse grid, reject outliers, and interpolate empty cells.
9. Map artificial-coral footprints, tank walls, inlet coordinates, and designated deposition regions.

10 Save the field, obstacle map, source zones, uncertainty estimates, and metadata as one site-model profile.

Do not change water depth, pump position, pump setting, major coral placement, lighting, or camera pose after calibration. If any changes, create a new profile.

Generate virtual-particle transport maps from this profile before the live run. Then perform a small set of labeled physical releases that were not used to choose model parameters. Those releases evaluate whether the calculated paths and deposition predictions match reality. The production concept replaces tank measurements with an appropriate local environmental data package; it does not require manually dropping plastic to build every source path.

7.5 Serial command contract

Use newline-delimited JSON or a compact framed protocol with acknowledgements. A readable hackathon contract is:

```
{"cmd":"HOME","request_id":"req_101"}
{"cmd":"MOVE_ABS","request_id":"req_102","x_mm":182.0,"y_mm":94.0,"speed_mm_s":35.0}
{"cmd":"PUMP","request_id":"req_103","on":true,"max_ms":2500}
{"cmd":"STOP","request_id":"req_104"}
```

Example telemetry:

```
{"type":"STATE","request_id":"req_102","state":"MOVING","x_mm":120.4,"y_mm":94.0}
{"type":"DONE","request_id":"req_102","x_mm":182.1,"y_mm":93.8}
{"type":"FAULT","code":"X_LIMIT","message":"Unexpected X limit hit"}
```

Rules:

- Commands are idempotent by `request_id` where possible.
- The Arduino rejects motion before homing.
- FastAPI sends one motion command at a time.
- A heartbeat timeout stops motion and turns the collection pump off.
- `STOP` and the physical emergency stop always take priority.

7.6 FastAPI interface

Recommended endpoints:

Method and path	Purpose
`POST /api/runs`	Start a calibrated run
`POST /api/runs/{id}/stop`	End or abort a run
`GET /api/runs/{id}`	Current state and summary
`GET /api/runs/{id}/tracks`	Track points and quality
`GET /api/runs/{id}/motion-state`	Moving, stationary, or uncertain classification
`GET /api/runs/{id}/estimate`	Inference mode, ranked inlets, confidence, evidence, Unknown reason
`POST /api/models/{id}/simulate`	Generate or refresh source-to-transport ensembles
`GET /api/models/{id}/coverage`	Current-field and deposition coverage summary
`POST /api/actions/{id}/approve`	Approve an unexpired collection action
`POST /api/actions/{id}/deny`	Deny and record reason
`POST /api/hardware/home`	Home in manual/admin mode
`POST /api/hardware/stop`	Immediate software stop

Method and path	Purpose
`GET /api/health`	Camera, database, serial, calibration, adapter health
`WS /api/events`	Live detections, flow, state, telemetry, and logs

The approval endpoint must revalidate target freshness, calibration ID, system state, and hardware health before moving.

8. Math, inference, and scientific limitations

8.1 Object position and velocity

For a filtered tank-coordinate track `p_k = [x_k, y_k]` at time `t_k`, use a centered difference when possible:

```
v_k = (p_(k+1) - p_(k-1)) / (t_(k+1) - t_(k-1))
```

At track boundaries, use a one-sided difference. Smooth positions with a Kalman filter, moving polynomial, or short Savitzky-Golay window before differentiation because derivatives amplify pixel noise.

The displayed speed is:

```
speed_k = sqrt(vx_k^2 + vy_k^2)
```

Report units in cm/s, not pixels/frame.

8.2 Measured surface-flow field

Let the measured surface velocity be:

```
u(x, y) = [u_x(x, y), u_y(x, y)]
```

Build `u` from reusable floating-marker trajectories at the fixed pump condition. In each spatial cell, calculate a robust mean or median velocity and a variance. Interpolate only across nearby populated cells. Mark poorly observed regions as low coverage instead of inventing vectors.

The field is an empirical surface-motion map for this tank configuration. It is not a full three-dimensional fluid simulation.

8.3 Motion-state classification

Over an observation window `T`, calculate filtered speed and net displacement:

```
mean_speed = mean(||v_k||)
displacement = ||p_end - p_start||
```

Classify the object as stationary only when both values remain below calibrated thresholds for enough consecutive frames. Use hysteresis so camera jitter does not cause rapid switching. If the evidence is inconsistent - for example, a plastic piece alternately catches and releases from coral - label it uncertain or segment the track into moving and deposited phases.

8.4 Moving-object inference: backward ensembles

For a moving object, ReefTrace has a trajectory rather than only an endpoint. Instead of drawing one reversed line, simulate an ensemble that perturbs the measured current field, release time, diffusion, and object drift. A simplified step is:

```
x_(n-1) = x_n - [u(x_n, t_n) + drift] * delta_t + noise
```

Score each candidate inlet by how well simulated paths originating there explain the observed positions, directions, and timestamps. Stop a trajectory when it reaches a candidate source zone, leaves the modeled domain, or exceeds the allowed release-time window.

This pathway usually produces stronger evidence because it uses multiple observed positions and velocities. Recirculation, wall effects, uncertain drift, and incomplete tracks still require an ensemble and an Unknown option.

8.5 Stationary-object inference: forward deposition ensembles

If the first reliable observation is already stationary, there is no observed velocity to reverse. For every candidate inlet `j`:

1. sample plausible release times within a configured window;
2. release `N` virtual particles from the source zone;
3. move them forward through the measured current field with uncertainty;
4. treat mapped coral or shoreline regions as absorbing or probabilistic trapping zones; and
5. measure how many simulated particles become deposited within radius `r` of the observed endpoint `x_star`.

The endpoint support for source `j` can be summarized as:

```
arrival_support(j) = arrivals_from_j_near_x_star / N
```

For faster operation, precompute a source-to-deposition transport kernel for each site model and query it when stationary plastic is detected. A long or unknown release-time window makes the result less specific. If several sources frequently feed the same accumulation zone, ReefTrace should rank them closely or return Unknown.

8.6 Physical trials are validation, not path training

The simulation engine should be generated from the measured current field, boundaries, source zones, obstacle map, and material assumptions. Labeled tank releases are then held out to answer:

- Does the moving-path ensemble contain the observed trajectory?
- Does the stationary deposition map predict the observed trapping region?
- How often is the top-ranked inlet correct?
- When the model is wrong or out of distribution, does it abstain?

The outside product follows the same pattern: generate the transport model from site data, then validate it with independent observations or drifters appropriate to that environment. It does not require manually teaching every location with plastic.

8.7 Unified candidate scoring

For moving plastic, candidate support is based on the likelihood of the observed trajectory under each inlet's backward/forward ensemble. For stationary plastic, candidate support is based on arrival and trapping density near the detected endpoint. Combine this support with explicit source priors only if those priors are documented:

```
source_support(j) = observation_likelihood(j) * source_prior(j)
match(j) = source_support(j) / sum_k source_support(k)
```

If no defensible prior exists, use equal priors. Unless the values are calibrated against held-out outcomes, label them **relative match scores**, not probabilities. Always display `inference_mode`, `model_version`, data coverage, release-time window, and the best-vs-second margin.

8.8 Unknown / abstention rule

Return Unknown if any of these conditions holds:

- track duration is below the minimum, such as 1.5-2.0 seconds;
- too many frames are missed;

- camera-to-tank calibration is stale or invalid;
- the track traverses a low-coverage flow region;
- the motion state remains uncertain;
- the stationary object's plausible release-time window is too broad;
- the deposition model lacks coverage around the observed endpoint;
- the best candidate's absolute fit is worse than threshold;
- the gap between the best and second-best candidates is too small;
- the predicted origin lands away from all inlet capture regions;
- the pump state or water depth differs from the active calibration profile.

Tune the absolute-fit and margin thresholds on held-out runs. The Unknown option is a correctness feature, not a failure.

8.9 Interception planning

For a short horizon, predict the target using the local measured flow or filtered object velocity:

```
x_target(t + tau) = x_now + integral_0^tau u(x(t)) dt
```

Choose `tau` to include expected gantry travel time. Reject the plan if the point is outside the safe workspace, too near an obstacle, older than the approval timeout, or associated with a low-quality track. Recompute immediately before motion.

8.10 Verification

After suction:

- search for the target in a radius around the collection point and along its short predicted path;
- require absence for several consecutive frames;
- optionally detect the object inside a camera-visible collection chamber;
- report `captured`, `not_captured`, or `uncertain`.

Do not label a target captured solely because the pump ran.

8.11 Limitations to state clearly

- Only visible floating objects are detected.
- Brightness, reflections, occlusion by the rover, and similar-colored decorations can affect detection.
- The flow map is valid only for the calibrated tank configuration and pump setting.
- Surface plastic may move differently depending on shape, buoyancy, wind from fans, and contact with walls.
- A stationary endpoint contains less information than an observed trajectory and may be consistent with several sources.
- Deposition predictions depend on simplified trapping rules for coral and boundaries; complex real habitat interactions require separate validation.
- Three candidate inlets simplify the inference problem; the true source in an open environment may be unlisted.
- Confidence reflects the model, calibration, and test distribution. It is not proof of causation.
- Tank-scale inlet attribution does not validate ocean, river, harbor, or geographic attribution.
- Real field deployment would require georeferenced sensors, time-varying current and wind fields, material-specific transport, release-time uncertainty, and independent validation.

9. Data model and database schema

Treat this as a logical schema that can be implemented in Butterbase or locally. Keep high-rate raw video and frame images on local disk; store references and summaries in the database.

site_models

Field	Type	Notes
id	UUID/text	Primary key
name	text	Tank or future field-site identifier
created_at	timestamp	UTC
domain_geometry_uri	text	Boundaries and obstacle map
source_zones_json	JSON	Candidate inlets or geographic source regions
current_field_uri	text	Measured or imported vector field
deposition_map_uri	text	Precomputed source-to-endpoint kernels
time_coverage	JSON	Valid timestamps or steady-state condition
uncertainty_json	JSON	Velocity, diffusion, drift, release-time assumptions
model_version	text	Reproducibility
validated	boolean	Independent validation completed

calibration_profiles

Field	Type	Notes
id	UUID/text	Primary key
created_at	timestamp	UTC
camera_hash	text	Camera and settings identifier
homography	JSON	3 x 3 image-to-tank matrix
gantry_transform	JSON	tank-to-gantry affine matrix
water_depth_cm	number	Measured depth
pump_setting	text/number	Locked run setting
flow_grid_uri	text	Local or object reference
mean_map_error_mm	number	Validation error
max_map_error_mm	number	Validation error
active	boolean	One active profile

inlets

Field	Type	Notes
id	text	A, B, or C
label	text	Human-readable name
x_cm, y_cm	number	Tank coordinates
capture_radius_cm	number	Source match region
active	boolean	Candidate enabled

runs

Field	Type	Notes
id	UUID/text	Primary key
started_at, ended_at	timestamp	Run interval
calibration_id	foreign key	Exact physical profile
site_model_id	foreign key	Transport model version
mode	enum	live, replay, manual
state	enum	State-machine state
actual_inlet	nullable text	Hidden ground truth, revealed after trial
final_outcome	enum	captured, missed, uncertain, aborted
notes	text	Operator notes

tracks

Field	Type	Notes
id	UUID/text	Track ID
run_id	foreign key	Parent run
object_label	text	visible plastic
started_at, ended_at	timestamp	Track interval
quality	number	0-1
point_count	integer	Observations
motion_state	enum	moving, stationary, uncertain
stationary_since	nullable timestamp	Deposition observation
trajectory_uri	text	Compressed points or file reference
mean_speed_cm_s	number	Summary

track_points

Store at a reduced rate, such as 5-10 Hz, rather than every camera frame.

Field	Type	Notes
track_id	foreign key	Composite index with time

Field	Type	Notes
t_ms	integer	Time from run start
x_cm, y_cm	number	Position
vx_cm_s, vy_cm_s	number	Velocity
detector_score	number	Vision quality
flow_x_cm_s, flow_y_cm_s	number	Local measured flow

source_estimates

Field	Type	Notes
id	UUID/text	Estimate ID
run_id, track_id	foreign keys	Evidence source
created_at	timestamp	UTC
site_model_id	foreign key	Transport model used
result	enum	A, B, C, Unknown
inference_mode	enum	moving_backward, stationary_forward, uncertain
score_a, score_b, score_c	number	Relative or calibrated scores
confidence_kind	enum	relative_match, calibrated_confidence
best_margin	number	Best minus second best
quality	number	Combined data quality
unknown_reason	nullable text	Abstention rationale
model_version	text	Reproducibility
evidence_json	JSON	Component distances and flags
release_window_json	JSON	Assumed release-time range
ensemble_summary_uri	text	Simulation counts and endpoint/path distributions

collection_actions

Field	Type	Notes
id	UUID/text	Action ID
run_id, track_id	foreign keys	Target
created_at, expires_at	timestamp	Approval window
target_x_cm, target_y_cm	number	Planned intercept
state	enum	proposed, approved, denied, executing, complete, fault, expired
requested_by	text	System/operator
hardware_request_id	text	Serial correlation
pump_ms	integer	Bounded suction duration
verification	enum	captured, not_captured, uncertain

approvals

Field	Type	Notes
id	UUID/text	Approval ID
action_id	foreign key	Collection action
channel	enum	dashboard, photon, manual
decision	enum	approved, denied
actor	text	Authenticated identity or demo operator
created_at	timestamp	UTC
message_id	nullable text	External correlation ID
reason	nullable text	Denial or note

system_events

Field	Type	Notes
id	UUID/text	Event ID
run_id	nullable foreign key	Parent run
timestamp	timestamp	UTC
level	enum	info, warning, fault
component	text	vision, api, arduino, adapter
code	text	Stable machine-readable code
payload	JSON	Sanitized details

Recommended indexes: `site_models.model_version`, `runs.started_at`, `tracks.run_id`, `tracks.motion_state`, `track_points(track_id, t_ms)`, `source_estimates(run_id, inference_mode)`, `collection_actions.run_id`, and `system_events(run_id, timestamp)`.

10. Dashboard design

10.1 Live operations screen

The live screen should fit on one laptop without scrolling:

- **Center:** camera feed with tank boundary, inlet labels, plastic bounding box, recent path, flow arrows, nozzle position, and planned intercept.
- **Top status bar:** camera, Arduino, calibration, pump setting, database, and emergency-stop status.
- **Mode banner:** `MOVING - TRAJECTORY INFERENCE`, `STATIONARY - DEPOSITION INFERENCE`, or `UNCERTAIN - OBSERVING`.
- **Right evidence panel:** ranked inlets, score type, confidence/match score, best-vs-second margin, current-field coverage, release-time window, and Unknown reason.
- **Transport layer:** toggle between observed path, backward ensemble, forward source particles, and deposition heatmap.
- **Action card:** target age, predicted intercept, approve, deny, and software stop.
- **Bottom timeline:** detected -> estimating -> awaiting approval -> moving -> suction -> verifying -> result.

10.2 Evidence design

Avoid a decorative single percentage. Show why the system made its choice:

```
Estimated source: Inlet B
Calibrated confidence: 0.81
Inference mode: moving - trajectory-conditioned ensemble
Track quality: High
Evidence:
  - 72% of retained backward ensemble reaches B's source zone
  - Observed motion fits the local field within tolerance
  - Best/second score margin: 0.19
  - Current-field coverage along track: 92%
Limit: attribution is among three known tank inlets only
```

If abstaining:

```
Estimated source: Unknown
Inference mode: stationary - forward deposition ensemble
Reason: Inlets A and C both feed this deposition region
Next step: widen observations, add temporal evidence, or retain Unknown
```

10.3 Calibration screen

- Four-corner camera calibration and validation errors.
- Gantry alignment targets and residual errors.
- Water depth and pump setting confirmation.
- Flow coverage heatmap and vector field.
- Mapped candidate sources, coral obstacles, and deposition regions.
- Virtual-particle count, release-time window, perturbation ranges, and model version.
- Activate/deactivate calibration profile.

10.4 History and evaluation screen

- Recent runs and outcomes.
- Confusion matrix for known-inlet trials.
- Separate metrics for moving and stationary cases.
- Transport-model coverage and validation results.
- Unknown rejection results.
- Collection success rate.
- Median detection-to-estimate and approval-to-capture time.
- Links to replay a stored trajectory.

10.5 Manual and safety screen

- Home, jog, pump pulse, stop, and rehome controls.
- Visible current coordinates and limit-switch states.
- Commands disabled when health or calibration checks fail.
- A permanent software stop control on every screen.

11. Sponsor integrations with claim discipline

The team should confirm the exact hackathon account, SDK version, credentials, quotas, and allowed demo flow before promising any integration. Product names are not architecture guarantees.

11.1 Chalk

Verified public documentation describes Chalk as a feature platform with features, resolvers, and online queries. A safe ReefTrace use is to expose derived, non-safety-critical context such as:

- `track.duration_s`
- `track.quality`
- `track.motion_state`
- `track.mean_speed_cm_s`
- `flow.coverage_ratio`
- `model.inference_mode`
- `model.release_window_s`
- `source.best_score`
- `source.best_margin`
- `system.collection_success_rate_1h`

Possible demo: FastAPI queries or writes these features, then displays them in the evidence panel or uses them as inputs to the deterministic estimator. The authoritative scores should still be calculated by versioned Python code. Do not claim Chalk performs computer vision, source attribution, robot control, or approvals unless the team implements and verifies those exact behaviors.

Fallback: compute the same feature dictionary locally and keep the rest of the system unchanged.

11.2 Butterbase

Verified public documentation describes an auto-generated REST data API after schema creation. ReefTrace can use it as the system of record for runs, estimates, approvals, actions, and summaries.

Implementation rules:

- Call Butterbase only from FastAPI; keep service credentials server-side.
- Buffer writes locally so a network outage cannot stop collection.
- Use idempotency keys or stable IDs to avoid duplicate events.
- Store raw video locally and persist only references and summary points remotely.
- Validate the exact schema, authentication mode, and row-level access before the demo.

Fallback: SQLite or append-only JSON locally, with a later sync.

11.3 Photon

Photon's current public product materials describe unified messaging for agents across supported communication channels and mention human-in-the-loop controls. A safe ReefTrace integration is an **approval transport**, not the owner of safety or motion:

1. FastAPI creates a short-lived collection action.
2. The Photon adapter sends a summary with run ID, target location, source estimate, confidence kind, and expiry.
3. An authenticated human approves or denies through a supported channel.
4. FastAPI validates the response, current state, target freshness, and hardware health.
5. Only then does FastAPI send a bounded command to Arduino.

Do not assume every channel supports buttons, callbacks, group messaging, identity mapping, or the same interaction model. Use the verified SDK's actual message/action primitives. If the required interaction is not available, send a notification through Photon and complete approval in the dashboard.

Fallback: the dashboard approval is the canonical path.

11.4 Sponsor integration claims table

Product	Safe claim after implementation	Claim to avoid
Chalk	"We serve and inspect derived context features used by our decision layer."	"Chalk identifies pollution sources."
Butterbase	"We persist structured run and outcome records through its supported data interface."	"Butterbase guarantees our real-time control loop."
Photon	"We route a human approval request through a verified supported channel."	"Photon autonomously controls the robot."

Official references checked

- Chalk documentation: https://docs.chalk.ai/docs/overview
- Butterbase REST API: https://docs.butterbase.ai/sdks-and-tools/rest-api/
- Photon product and messaging overview: https://photon.codes/
- Photon 2.0 concepts: https://photon.codes/blog/introducing-photon-2.0

These references confirm general product positioning as of this plan; the team must still verify hackathon-specific access and exact SDK behavior.

12. Thirty-six-hour build plan

Hours 0-3: Freeze scope and de-risk

- Pin the source-first thesis above the workspace: attribution is the product; collection is the response layer.
- Confirm both required inference scenarios: moving and stationary/deposited plastic.
- Inventory every component, voltage, cable, driver, and credential.
- Assemble the empty tank and rigid overhead frame.
- Create the repository, environment files, issue board, and API contracts.
- Record one short sample video for development even before the robot is ready.

Gate: camera sees the entire tank; Arduino connects; both pumps and both motors have verified power requirements.

Hours 3-8: Hardware baseline and manual control

- Assemble and square the gantry.
- Wire limit switches, stepper drivers, MOSFET, fuse, and emergency stop.
- Implement Arduino home, move, pump pulse, stop, and fault telemetry.
- Build a bare manual FastAPI or script control panel.
- Leak-test the collection loop.

Gate: click/manual coordinates can move the nozzle safely, and suction captures a test piece in at least 4 of 5 stationary trials.

Hours 5-11: Vision and motion-state baseline

- Lock camera and lighting.
- Implement tank mask, simple color/brightness segmentation, and tracking.
- Implement moving, stationary, and uncertain classification with hysteresis.
- Estimate and validate the image-to-tank homography.
- Emit live position and velocity over WebSocket.
- Save replay fixtures.

Gate: a visible test piece remains tracked for at least 90% of frames on a standard path, with mean coordinate error below 5 mm.

Hours 9-16: Current field and site digital twin

- Collect repeated surface-marker trials at the frozen pump setting.
- Generate the vector grid and coverage map.
- Map tank boundaries, artificial-coral obstacles, three source zones, and deposition regions.
- Implement virtual-particle transport with configurable velocity uncertainty and diffusion.
- Precompute forward source-to-deposition maps for all three inlets.

Gate: the engine can release virtual particles from each inlet, visualize their paths, and mark poorly covered regions without using labeled plastic paths as its source database.

Hours 14-22: Dual-mode source estimation and abstention

- Implement trajectory-conditioned backward ensembles for moving objects.
- Implement forward source-to-deposition ensembles for stationary objects.
- Add explicit release-time windows, source priors, ensemble perturbations, and evidence summaries.
- Collect labeled physical trials only after the model is frozen, then evaluate on those held-out runs.
- Add Unknown for poor fit, small margin, and low quality.
- Display relative match unless calibration supports stronger language.

Gate: both inference branches work, the dashboard names the branch used, and held-out trials show either a correct result or an honest abstention with no confidently wrong result in the frozen demo configuration.

Hours 18-25: Close the physical loop

- Fit tank-to-gantry transform.
- Implement interception and safe workspace checks.
- Add state machine and approval expiry.
- Move, suction, and verify on live tracks.
- Tune capture timing and nozzle placement.

Gate: the source-estimation demo remains complete even with collection disabled; when enabled, collection succeeds in at least 4 of 5 representative rehearsals.

Hours 22-29: Dashboard and persistence

- Build the live camera overlay and evidence panel.
- Add motion-mode banner, ensemble/path toggle, deposition heatmap, and model-assumption panel.
- Add safety/health status, approval card, stop control, and timeline.

- Implement the logical database schema in Butterbase or local fallback.
- Add history and run-replay views.

Gate: a new teammate can understand the current state and safely approve or stop without terminal access.

Hours 27-32: Sponsor adapters

- Implement Butterbase writes and local retry queue.
- Add the Chalk feature-context adapter if credentials and docs are confirmed.
- Add Photon notification/approval transport if the exact supported workflow is confirmed.
- Keep feature flags and health checks for all adapters.

Gate: disabling any sponsor adapter does not break the core loop.

Hours 30-35: Test, rehearse, and package

- Run the test matrix below.
- Rehearse one moving-source case and one stationary-source case.
- Save one known-good calibration profile and do not move the rig.
- Record a clean replay and a manual capture fallback.
- Rehearse the 3.5-minute script with a timer.
- Prepare a one-slide architecture diagram and one-slide scientific boundary.

Hours 35-36: Freeze

- No new features.
- Label cables, clean the tank, charge the laptop, and stage spare parts.
- Restart the complete system once.
- Run one final live trial and one fallback replay.
- Export results and protect the calibration files.

13. Team roles

Role 1 - Hardware and embedded lead

- Owns gantry, pumps, collection chamber, wiring, safety, Arduino, and spare parts.
- Has final authority to stop motion or disable the collection pump.
- Produces the hardware health checklist before every live demo.

Role 2 - Vision and inference lead

- Owns camera calibration, OpenCV detector/tracker, motion-state classifier, current field, forward/backward transport ensembles, inlet scoring, Unknown logic, and replay fixtures.
- Maintains the scientific claims and evaluation results.

Role 3 - Backend and integration lead

- Owns FastAPI, site-model versioning, simulation jobs, state machine, serial client, action/approval contracts, database schema, and sponsor adapters.
- Ensures external services cannot bypass local safety validation.

Role 4 - Frontend, testing, and presentation lead

- Owns Next.js, overlays, evidence design, run history, demo script, judge interaction, and test log.
- Runs rehearsals and keeps the presentation within time.

If the team has three people, combine backend with frontend. If it has two, one person owns physical systems and one owns software; postpone sponsor adapters until the local loop works.

14. Test and validation plan

14.1 Unit tests

- Homography converts known pixels to known tank points.
- Velocity uses real timestamps and handles dropped frames.
- Motion-state classification resists pixel jitter and changes mode only after the configured observation window.
- Flow interpolation returns low coverage outside supported cells.
- Forward and backward simulations are reproducible under a fixed random seed.
- Stationary inference queries deposition support rather than inventing a zero-velocity backtrack.
- Candidate scoring is deterministic for fixed inputs.
- Unknown triggers for short track, low fit, small margin, stale calibration, and changed pump setting.
- Tank-to-gantry transform rejects points outside the safe polygon.
- Approval rejects expired, duplicate, wrong-state, and stale-target actions.
- Serial parser handles partial lines, malformed messages, timeouts, and duplicate acknowledgements.

14.2 Hardware tests

- Emergency stop removes motor and pump power.
- Startup leaves pump and motors off.
- Every limit switch stops motion in the expected direction.
- Homing repeats within 2 mm.
- Driver and motor temperatures remain acceptable through the demo cycle.
- Tubing remains attached and leak-free for 20 minutes.
- Pump cutoff works at the configured maximum duration.
- Mesh chamber captures the largest and smallest test pieces.

14.3 Vision tests

- Static coordinate error at 5-9 validation points.
- Tracking through the center, near walls, and near coral reflections.

- False-positive trial with no plastic present.
- Brief rover occlusion and reacquisition.
- Lighting disturbance and recovery.
- Moving, stationary, intermittently stuck, and uncertain motion sequences.
- Multiple objects: either track both or explicitly reject the run if only one is supported.

14.4 Source-estimation experiment

Freeze the current field, source zones, trapping rules, perturbation ranges, and scoring thresholds before the labeled releases. Minimum credible hackathon set:

- Moving held-out tests: 3 labeled releases per inlet, 9 total.
- Stationary held-out tests: release from a known inlet and allow at least 6 pieces to become deposited before scoring the endpoint. Manually placed pieces may test UI and Unknown handling, but they do not validate source accuracy.
- Unknown tests: at least 3 observations from non-inlet positions, ambiguous deposition zones, or altered flow conditions.

The model may use reusable flow-marker observations to estimate the current field, but it must not memorize the labeled plastic-test paths. Record separate confusion matrices for moving and stationary modes, number of Unknown results, and any confident errors. Do not silently rerun or remove failed trials.

Targets, not guaranteed outcomes:

- At least 80% of held-out known-inlet trials correct or honestly abstained.
- Zero high-confidence wrong predictions in the frozen demo condition.
- At least 2 of 3 out-of-distribution tests rejected as Unknown.
- Stationary cases show lower or equal confidence than comparable moving cases unless the deposition evidence is genuinely more selective.
- Source estimate available within 2 seconds after the minimum track length.

If the sample is too small to calibrate probabilities, use relative match language.

14.5 End-to-end trials

Run at least five rehearsals using the complete sequence:

1. System health check.
2. Hidden inlet release.
3. Detection, tracking, and motion-state classification.
4. Moving or stationary inference with source estimate or Unknown.
5. Approval.
6. Rover motion.
7. Suction.
8. Verification.
9. Persistence and replay.

Target: 4 of 5 successful captures with no unsafe motion. Preserve one failed or Unknown example to show scientific honesty if asked.

Run at least two additional source-only rehearsals with the collector disabled. The source product must remain demonstrable even if the response hardware is unavailable.

14.6 Demo readiness checklist

- [ ] Camera, tank, coral, pump, and inlet geometry match the active calibration.
- [ ] Active site model includes boundaries, source zones, current field, deposition map, and uncertainty settings.
- [ ] Moving and stationary replay fixtures both pass.
- [ ] Water depth is at the calibration mark.
- [ ] Emergency stop and software stop tested.
- [ ] Gantry homed and safe polygon loaded.
- [ ] Collection chamber empty and visible.
- [ ] Test plastic counted.
- [ ] Local database writable.
- [ ] Sponsor adapter health known; fallbacks ready.
- [ ] Replay video and sample run available offline.
- [ ] No secrets visible in browser, logs, or terminal.

15. Live demo script

0:00-0:30 - The differentiated problem

> *"Ocean cleanup removes today's debris. ReefTrace addresses the upstream question: where is the plastic entering, and how certain are we?"*

0:30-0:55 - Site model and scope

Point to the tap water, artificial coral, three candidate inlets, fixed camera, circulation pump, and transport-map view.

> *"This tank is a localized digital twin. We measured its current field, mapped its boundaries and trapping regions, and calculated how virtual plastic can travel from each inlet. Physical releases test the model; they do not manually teach every path."*

0:55-1:20 - Two automatic inference modes

Show the decision branch in the dashboard.

> *"ReefTrace first decides what evidence exists. Moving plastic is analyzed from its observed trajectory. Stationary plastic is analyzed from where it was deposited and which simulated source paths can reach that location."*

1:20-1:50 - Moving scenario: hidden source

Ask a judge to choose A, B, or C and release one test piece. Keep the true inlet hidden from the software. Start the run.

1:50-2:20 - Moving inference

Show the bounding box, motion-state label, physical position, velocity vector, measured flow arrows, and backward ensemble.

> *"The system is not simply drawing the track backward. It tests an ensemble of uncertain transport paths against all three source zones."*

Show the ranked candidates, score type, evidence coverage, and Unknown logic. Ask the judge to reveal the inlet after the result is frozen.

2:20-2:55 - Stationary scenario

Place or reveal a second piece already resting on a mapped coral zone. The dashboard should classify it as stationary and switch modes automatically.

> *"This object has no observed velocity, so ReefTrace does not pretend it can reverse a missing path. It compares forward simulations from every inlet over a release-time window and shows which sources can plausibly feed this deposition region."*

Show the deposition heatmap, ranked sources, lower confidence if appropriate, and Unknown if multiple sources converge.

2:55-3:20 - Product transfer beyond the tank

Show the site-model inputs panel.

> *"For an outside deployment, the tank vector field is replaced by a local package of boundaries, candidate source zones, currents, and wind, tide, or wave data where relevant. The inference runs automatically once the site model is configured; we do not need to manually release plastic for every observation. Independent validation is still required before treating the results as geographic evidence."*

3:20-3:45 - Optional response layer

Show the collection action. Approve in the dashboard, or demonstrate Photon if the verified integration is healthy.

> *"Cleanup is not our main differentiator. It is the response layer: once ReefTrace has produced source intelligence, a human can authorize removal."*

The rover moves, suction runs, and the plastic appears in the mesh chamber. The camera verifies the target is absent.

3:45-4:00 - Close

Show the stored run and result.

> *"ReefTrace turns cleanup from a repeated reaction into source intelligence: observe, classify, model, attribute, explain uncertainty, and then respond."*

Questions the team should be ready for

Why not use real coral? It would be unethical and technically inappropriate for a freshwater hackathon test. Artificial coral provides repeatable geometry without harming an organism.

Why tap water? The demonstration needs controlled optics and repeatable flow. The water's origin does not create the current; the pump, boundaries, depth, and obstacles do.

Do you need to release plastic at every new location? No. A deployment requires a site transport model built from local geometry and environmental data. Physical observations or drifters validate that model; they are not a manual database of every possible path.

How can stationary plastic have an origin estimate? ReefTrace simulates transport forward from every candidate source over a release-time window and evaluates which sources can deposit plastic near the observed endpoint. This is weaker evidence than a full observed trajectory and may correctly return Unknown.

Is the confidence a probability? Only if calibrated against held-out trials. Otherwise the dashboard calls it a relative match score.

Can this locate a real polluter? No. It ranks three known tank inlets. Real geographic attribution requires field-scale data, material transport models, and validation, and still would not by itself identify a responsible company or person.

Why use AI? Computer vision extracts observations; the source estimate is mathematical and testable. Optional agent/context tools help explain evidence and route approvals, but they do not invent the confidence score.

What is novel? The central value is a dual-mode, site-adaptive source-attribution engine: trajectory-conditioned inference for moving debris and deposition-conditioned inference for stationary debris. Collection is an optional response and verification layer.

16. Fallback scope ladder

Choose the highest level that is reliable by hour 30. A polished smaller loop is better than an unstable larger one.

Level 1 - Full demonstration

- Live vision, motion-state classification, and measured flow overlay.
- Moving-object backward ensemble and stationary-object forward deposition ensemble.
- Three-inlet estimate with evidence, uncertainty, and Unknown.
- XY interception.
- Dashboard or Photon approval.
- Suction capture, verification, and persistence.

Level 2 - Full source intelligence, fixed collection station

If the gantry is unreliable, park the nozzle at a fixed capture station. Predict when the plastic will enter its zone, request approval, and activate suction. Preserve source estimation, verification, and data logging.

Level 3 - Full source intelligence, manual collection

If motor control fails, let a teammate position the nozzle while the software still performs detection, flow measurement, source estimation, approval, timed suction, and verification. Describe the mechanical failure honestly.

Level 4 - Live source inference plus replayed second scenario

If time or tank conditions prevent both scenarios from running live, demonstrate one mode live and replay raw footage for the other through the real inference pipeline. A manual suction capture may still show the response layer.

Level 5 - Source-intelligence evidence only

As a last resort, present recorded tank observations, the measured vector field, moving and stationary ensemble visualizations, held-out source-estimation results, and Unknown behavior. Collection can be omitted without changing the project's core claim. Do not simulate a successful live action and imply it happened.

17. Risks and mitigations

Risk	Early warning	Mitigation	Fallback
Reflections break detection	Centroid jumps or false blobs	Diffused side lighting, dark surround, locked exposure, high-contrast test plastic	Replay fixture
Flow is chaotic or inconsistent	Repeated marker measurements disagree	Lower pump speed, simplify coral placement, freeze depth and return flow	Use wider Unknown region
Transport model does not match physical motion	Held-out paths fall outside simulated ensemble	Increase measured coverage, revise uncertainty, check boundaries and timestamps	Report model invalid and use replay
Inlet paths overlap	Small score margins	Extend observation time, relocate inlets, use crossing gates	Honest Unknown
Stationary endpoint fits several sources	Similar deposition support	Show release-time assumptions and ranked alternatives	Unknown, no unique attribution

Risk	Early warning	Mitigation	Fallback
Field site lacks suitable current data	Large gaps or overly coarse resolution	Add local sensors or choose a better-supported pilot site	Do not claim site attribution
Camera or tank moves	Calibration error rises	Rigid mounts, taped feet, calibration checksum and quick validation	Recalibrate
Gantry misses target	Transform residual or belt slip	Rehome, lower speed, add approach margin, recalibrate G	Fixed nozzle
Suction pushes target away	Object accelerates near nozzle	Lower return jet, wider nozzle, approach upstream, tune pump pulse	Manual capture
Plastic clogs pump	Flow drops	Mesh chamber before pump, size limits, removable cleanout	Stop and clear unpowered
Water leak	Drips or falling level	Clamps, tray, leak test, raised electronics, E-stop	End physical demo
Serial or USB disconnect	Missing heartbeat	Watchdog, outputs off, reconnect and rehome	Manual mode
Sponsor API unavailable	Failed health check	Feature flags, local adapters, queued writes	Core local loop
Approval arrives late	Target moved	Short expiry, recompute before motion	New action required
Confidence is overstated	Tiny test set	Relative-score label, held-out trials, Unknown	Show evidence not percent
Multiple objects enter	Track association switches	One-object demo rule or reject ambiguous run	Reset trial
Team runs out of time	Core loop incomplete by hour 24	Freeze sponsor work and visual polish	Scope ladder

18. Presentation assets

Prepare only a few visuals; the live system is the centerpiece.

Slide 1 - Problem and product

- Reactive cleanup misses repeated entry points.
- ReefTrace is pollution source intelligence, not primarily another cleanup robot.
- One clear sentence on scope: visible floating plastic, three known tank source zones, two inference modes.

Slide 2 - Architecture

Show camera -> motion-state classifier -> site transport model -> moving/stationary inference -> ranked sources. Put optional approval and collection after the source result. Keep Chalk, Photon, and Butterbase outside the scientific and safety-critical paths.

Slide 3 - Evidence and honesty

- A moving backward ensemble and stationary deposition heatmap.
- Candidate ranking plus Unknown.
- Separate held-out results for moving and stationary cases.
- Large caption: "Tank inlet attribution is not geographic attribution."

Physical labels

Label Inlets A/B/C, circulation pump, fixed camera, above-water rover, suction nozzle, collection chamber, emergency stop, and water return. Good labels reduce verbal explanation.

19. Roadmap after the hackathon

Near term - improve the tabletop evidence

- Collect more held-out trials across plastic shapes and pump settings.
- Calibrate confidence with a larger dataset.
- Add explicit multi-object tracking.
- Estimate uncertainty bands for flow vectors, backward trajectories, and stationary deposition kernels.
- Improve trapping models for coral-like obstacles and intermittent sticking.
- Automate camera, flow, and gantry calibration checks.
- Compare fixed-station and moving-nozzle collection efficiency.

Medium term - larger controlled environments

- Test in a flume or wave tank with instrumented currents.
- Add wind measurements and material-specific drag parameters.
- Build an automated site-model importer for boundaries, candidate source zones, and time-indexed environmental fields.
- Use multiple cameras or depth sensing while keeping electronics dry.
- Evaluate obstacle-aware interception and longer-duration operation.
- Partner with environmental scientists to design validation protocols.

Long term - field research, not a direct scale-up claim

- Replace tank coordinates with georeferenced observations.
- Fuse local current sensors with suitable historical and real-time current, wind, tide, and wave products.
- Model uncertain release time and simulate ensembles rather than one path.
- Support moving, stationary, and intermittently deposited observations in one data-assimilation framework.
- Validate against controlled releases or documented events before reporting geographic confidence.
- Assess permitting, wildlife interaction, navigation, weatherproofing, maintenance, cybersecurity, and retrieval.
- Explore underwater or additional-material systems only as separate designs with appropriate sensors and transport models.

The field roadmap should begin with measurement and validation. It should not assume that the tabletop estimator transfers unchanged to a river, harbor, or ocean.

20. Final definition of done

ReefTrace is ready to present when all of the following are true:

- The physical setup matches a validated calibration profile.
- Visible floating plastic is detected and tracked in centimeters with acceptable error.

- The system classifies observations as moving, stationary, or uncertain using documented thresholds.
- Surface flow has been measured at the frozen pump setting and its coverage is shown.
- The active site model contains boundaries, candidate sources, obstacles, current field, uncertainty settings, and a version identifier.
- Moving objects use trajectory-conditioned backward ensembles.
- Stationary objects use forward source-to-deposition ensembles over an explicit release-time window.
- The estimator ranks A/B/C or returns Unknown using documented fit and separation thresholds.
- The dashboard distinguishes relative match from calibrated confidence.
- Physical source releases are used as held-out validation, not as the path-training database.
- The source-attribution demonstration remains complete with collection disabled.
- The rover or approved fallback physically captures plastic as the optional response layer.
- A human approval is required before collection.
- Arduino enforces homing, limits, bounded pump time, watchdog, and faults.
- The camera reports captured, not captured, or uncertain after the action.
- The run is stored locally, with Butterbase sync if healthy.
- Chalk, Photon, and Butterbase can each be disabled without breaking the core loop.
- The team can state the scientific boundary in one sentence:

> *ReefTrace demonstrates dual-mode source attribution for moving and stationary plastic among known inlets in a controlled tank; real geographic attribution requires a separate site model and field validation.*

That is a focused, credible, and compelling 36-hour project.